BATCH5_RTA_GEOM

PREREG: BATCH5_RTA_GEOM (Map-C19, Map-C20): fibre-by-fibre permanent-class survival

Written 2026-10-24 at about 7:10 PM CDT, before any BATCH5 test. Owner: Arvin Hampton. New files only. CORE_FREEZE. All maps are **Category B**; a pass is a survived prediction, not a bridge. order20 and the RTA_FOUNDATION_BOUND paper are not touched. No Π, grading or path predicate is invented. C4 and C15–C18 are not reopened. Runs: nice, OMP_NUM_THREADS=1, ulimit -v 3000000.

Written survival criterion (quoted from the source; not invented)

/tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/scripts/residual_p_plus_multiscale_probe.py, lines 139–184. On a set of 2-cells (charge triangles T(a,b,c;i) and squares S(a,b;i), a < b < c), form the vectors: - $t_w = \omega_2$ on triangles (0 on squares); - $s_{ad} = \alpha(a,b)\cdot\delta f(i)$ on squares (0 on triangles); - $t_{wf} = \omega_2\cdot f(i)$ on triangles; - $\eta = s_{ad} - t_{wf}$.

The class survives iff $s_{ad} \notin \mathrm{span}(t_w, \eta)$ (the probe: lstsq residual > 1, with rank[t_w, s_{ad}, t_{wf}] = 3 and rank[t_w, η] = 2). Here $\omega_2(a,b,c) = \mathrm{sgn}(a-b) + \mathrm{sgn}(b-c) + \mathrm{sgn}(c-a)$, $\alpha = \mathrm{sgn}(a-b)$, $f(i) = \tau(x_i)$, $\delta f(i) = f(i+1) - f(i)$, $x_j = 21+9j$, τ as in BATCH4.

Fibre slice (interpretation fixed now; the source has no per-fibre version)

Slice r (r = 0..7) is all triangles T(a,b,c;i) with $i \equiv r \pmod 8$, $i \in 0..538$, together with all squares S(a,b;i) on path edges e_i with $i \equiv r \pmod 8$, $i \in 0..537$. All a < b (< c) in 0..8 are used, i.e. all cells, not the probe's subsample. Exact test: integer Gram determinants. Survive_r ⇔ rank[t_w, η, s_{ad}] = 3 and rank[t_w, η] = 2. Full-carrier control: the same criterion on all i must give survival (it reproduces the written lock); otherwise the run is VOID.

Map-C19 (primary; datum: C15 measured A2c full support, k = 105 of 105)

Prediction: full support on the modular side ⇒ the class survives in **all 8** fibres. PASS iff Survive_r for every r = 0..7. Low discrimination is disclosed: C20 is the discriminating rival, and at most one of the two can pass.

Map-C20 (discriminating; datum: discriminant 140, Kronecker character χ_{140})

Prediction: the class survives in fibre r **iff** $\chi_{140}(2r+1) = +1$ (odd lift, since χ_{140} vanishes on even arguments). PASS iff the survival set equals $\{r : \chi_{140}(2r+1) = 1\}$ exactly.

Still OPEN (blocked, not re-hashed): C6, C10, C11, E1–E3.

RESULTS: BATCH5_RTA_GEOM

PREREG.md sha256 35563f1388e39795799b6f37b0f7987aa2c1add25677d1dfb9e0160688c48379, hashed before tests. Script scripts/b5.py (hash in scripts/SHA256SUMS); data data/b5.json; log logs/b5.log. Exact integer Gram ranks (sympy), under 1 s.

Test	Measured	Verdict
Full-carrier control (written lock)	rank[t_w, η] = 2, rank[t_w, η, s_ad] = 3 → survives	control OK (run valid)
Map-C19: survives in all 8 fibres	survival set {0, …, 7}	**PASS** (Category B), **non-discriminating**, see below
Map-C20: survives iff $\chi_{140}(2r+1) = 1$, predicted set {0, 4, 6}	survival set {0, …, 7}	**FAIL**
C6, C10, C11, E1–E3	still blocked	OPEN

Why the C19 pass carries no modular content (Category A, elementary; found after the run, verdict unchanged)

- On any slice, suppose s_ad = x·t_w + y·η. The squares force y = 1, if s_ad ≠ 0 on the slice. The triangles then force $x \cdot \omega_2 = \omega_2 \cdot f(i)$ for every slice vertex i, i.e. f is constant on the slice.
- Hence **survival on a slice ⇔ (δf ≠ 0 on some slice edge) and (f not constant on the slice vertices).**

- Every fibre has 26–40 tower jumps (BATCH4 C16) and non-constant f, so survival in all 8 fibres is forced by the carrier alone, whatever the modular datum.
- The C19 “pass” is therefore a survived but vacuous prediction. It is **not** a Residual–Resonant or geometric bridge, and no bridge claim is made.
- The same lemma shows that any per-fibre (or any sub-sampled) version of this criterion is non-discriminating. The written thin-F survival test cannot single out fibres. # NOTES: BATCH5_RTA_GEOM ## Sources read (cited)
- /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/P_plus_Permanent_Class_Survival_Optional_Lock_2026-08-08.md (statement: survival in thin F; H^2 proxy rank 1; 56 same-tower triples). Copies are in oct17-master/docs/…, reports/self-audit/repo/docs/… and reports/bridges/_git_v8_unpushed_clone/docs/….
- /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/Residual_Product_Complex.md §§1–5 ($X_{\times}$, α, ω_2, η, $D\eta = 0$).
- /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/Thin_Complex_Formality.md §1 (generators, relations, $H^2(F) = Q\cdot[\alpha\otimes\delta f]$).
- /tmp/repo539/docs/S2-11DM2ET-X-Axiomatic-Book/scripts/residual_p_plus_multiscale_probe.py lines 139–184: the operational survival criterion used verbatim.
- Fibre and tower labelling: MAPS_CANDIDATES/BATCH3_RTA_GEOM/NOTES.md, BATCH4_RTA_GEOM/NOTES.md; reports/crypto/RPP_spec_v1/RESIDUAL_T3_predicate_sources.md §2.
- Modular datum for C19: BATCH4 C15 ($k = 105$). For C20: disc 140 of $Q(\sqrt{35})$.

Disclosures

- The per-fibre slice (vertices and edges with $i \equiv r \bmod 8$) is an interpretation fixed in the PREREG. The sources define survival only on the full carrier.
- Category: the C19/C20 verdicts are B. The equivalence lemma in RESULTS is Category A (two lines of linear algebra).

Sharpest next step

- The thin-F survival criterion is too coarse to tie to any modular datum: it detects only “f non-constant and $\delta f \neq 0$”.
- A discriminating cohomological map needs a finer written invariant. Candidates in the sources: the **56 same-tower triples of P^+** (written; 3 in window $W = 18$, 53 in the tower region) and the **P^+ chord-zero path-H^2 proxy** (probe lines 200–213, $56 - \mathrm{rank}\, dP_cz$).
- Next prereg: predict the per-fibre distribution of that written path-H^2 proxy, or of the 56 triples, from a fixed modular datum. Example: the P-odd/P-even 51/54 split, or RG1′ zero-set structure. All of these must be fixed before computing.
- Otherwise, the Resonant maps stay blocked on a written Π or ± 1 grading.
- order20 and the paper stream were not touched. CORE_FREEZE; new files only; no push, no sends.